the distant Sun. Turn off the Sun, and they would be only a little affected.

But Uranus is another story. Uranus is an anomaly among the Jovian planets. Uranus is like the Earth: There's very little intrinsic heat pouring out. We have no good understanding of why this should be, why Uranus—which in many respects is so similar to Neptune—should lack a potent source of internal heat. For this reason, among others, we cannot say we understand what is going on in the deep interiors of these mighty worlds.

Uranus is lying on its side as it goes around the Sun. In the 1990s, the south pole is heated by the Sun, and it is this pole that Earthbound observers at the end of the twentieth century see when they look at Uranus. It takes Uranus 84 Earth years to make one circuit of the Sun. So in the 2030s, the north pole will be sunward (and Earthward). In the 2070s the south pole will be pointing to the Sun once again. In between, Earthbound astronomers will be looking mainly at equatorial latitudes.

All the other planets spin much more upright in their orbits. No one is sure of the reason for Uranus' anomalous spin; the most promising suggestion is that sometime in its early history, billions of years ago, it was struck by a rogue planet, about the size of the Earth, in a highly eccentric orbit. Such a collision, if it ever happened, must have worked much tumult in the Uranus system; for all we know, there may be other vestiges of ancient havoc still left for us to find. But Uranus' remoteness tends to guard its mysteries.

In 1977 a team of scientists led by James Elliot, then of Cornell University, accidentally discovered that, like Saturn, Uranus has rings. The scientists were flying over the Indian Ocean in a special NASA airplane—the Kuiper Airborne Observatory—to witness the passage of Uranus in front of a star. (Such passages, or occultations as they're called, happen from time to time, precisely because Uranus slowly moves with respect to the distant stars.) The observers were surprised to find that the star winked on and off several times just before it passed behind Uranus and its atmosphere, then several times more just after it emerged. Since the patterns of winking on and off were the same before and after